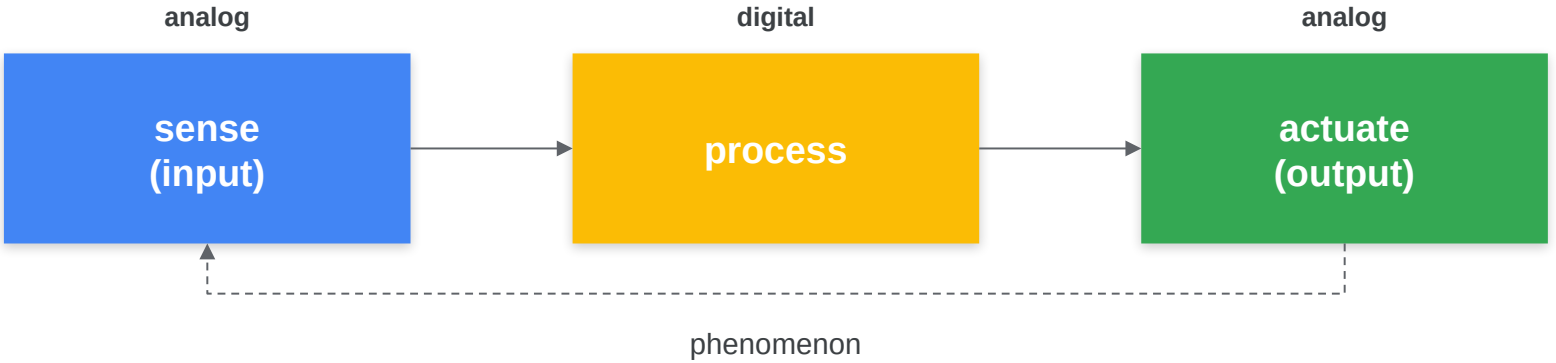
Embedded System



System Elements

We start with a simple question: **what is an embedded system?**



Why ***embedded***?

definition: “(adj) fixed firmly and deeply in a *surrounding mass*; implanted.”


10:01


nest
nest
22
TURN OR PUSH
TO DEMO



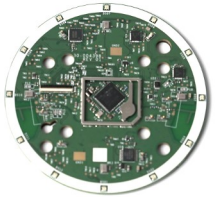






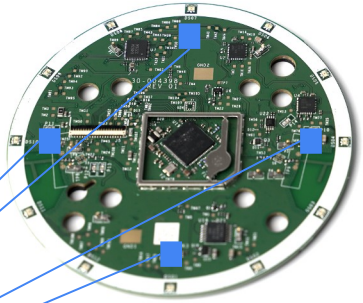






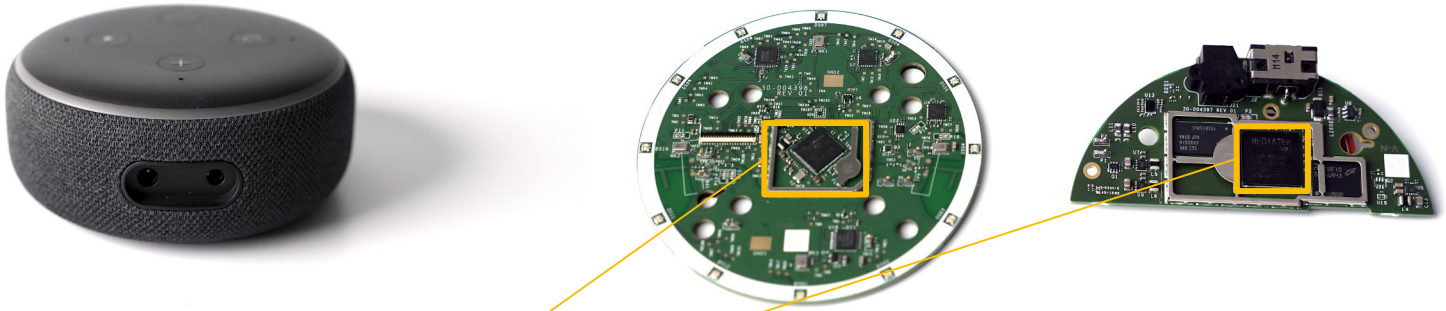








sense
(input)

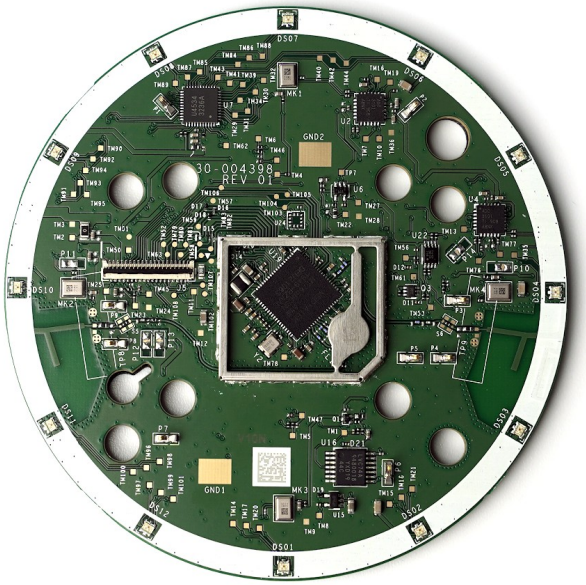


process

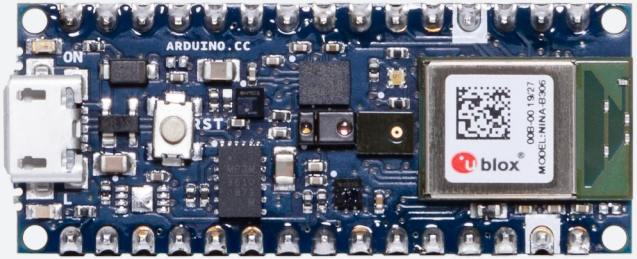




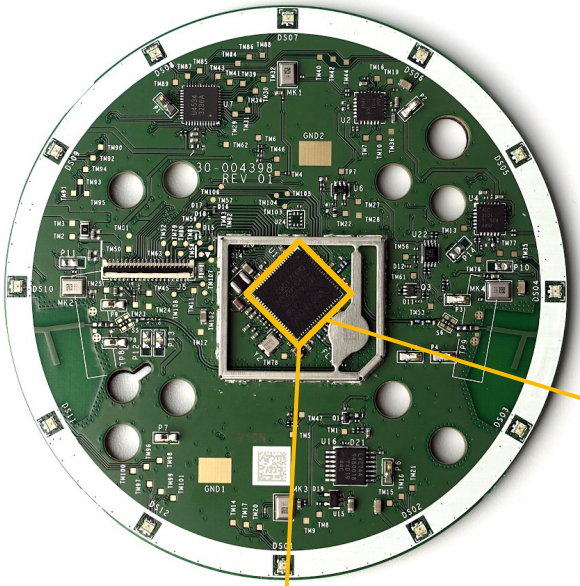
actuate (output)



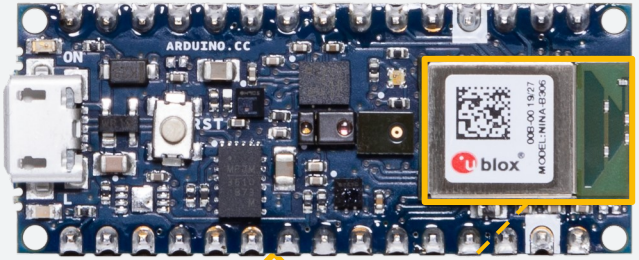
Your Embedded System

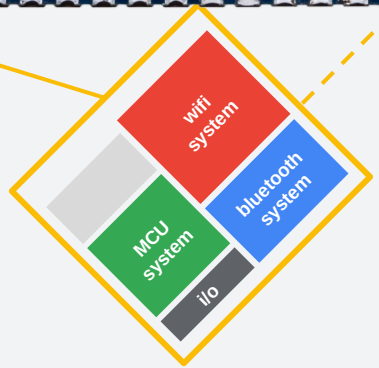


Your Embedded System

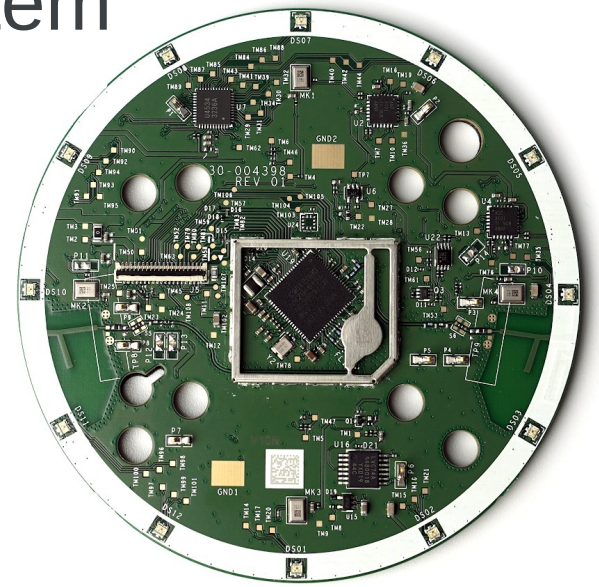


MediaTek 7658CSN: Wi-Fi +ARM® Cortex-R4

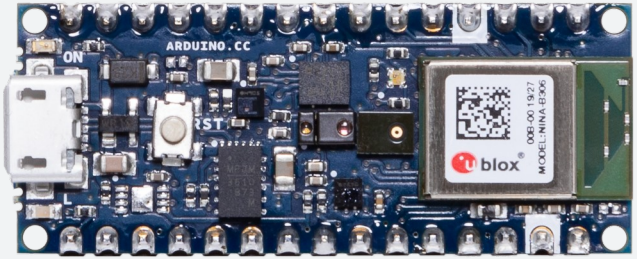


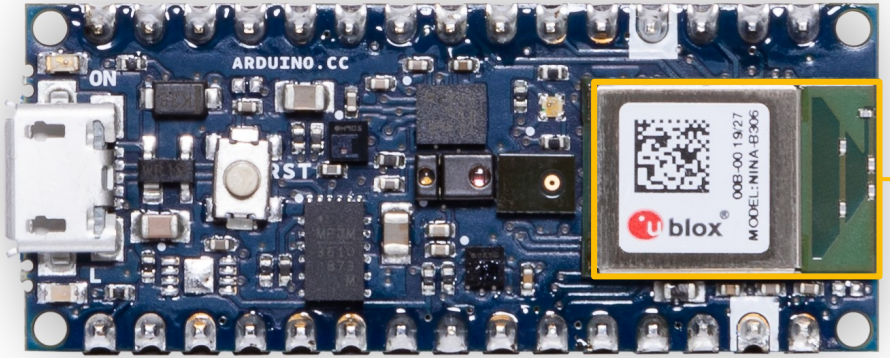


Real Embedded System

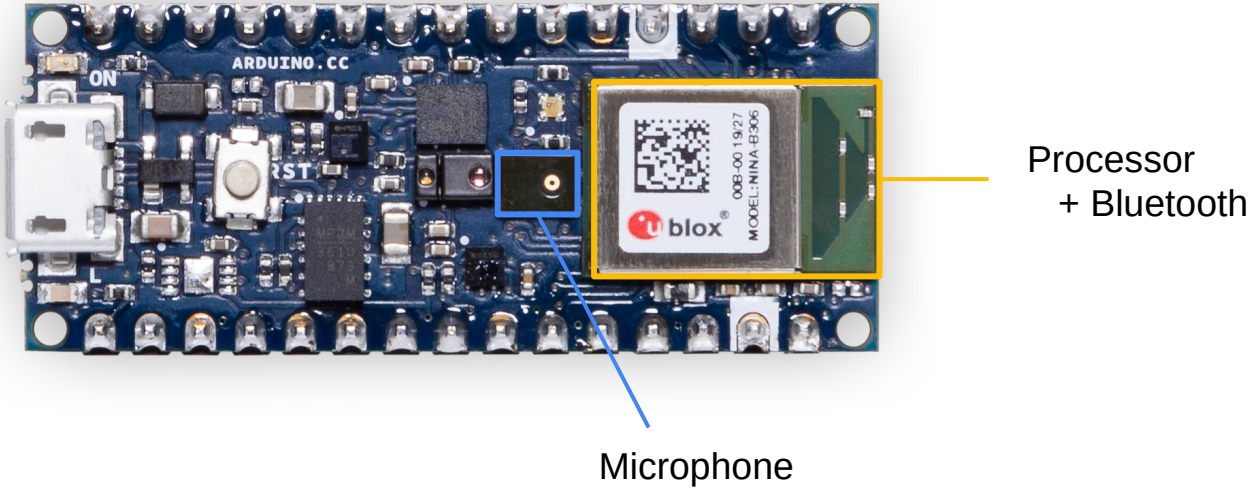


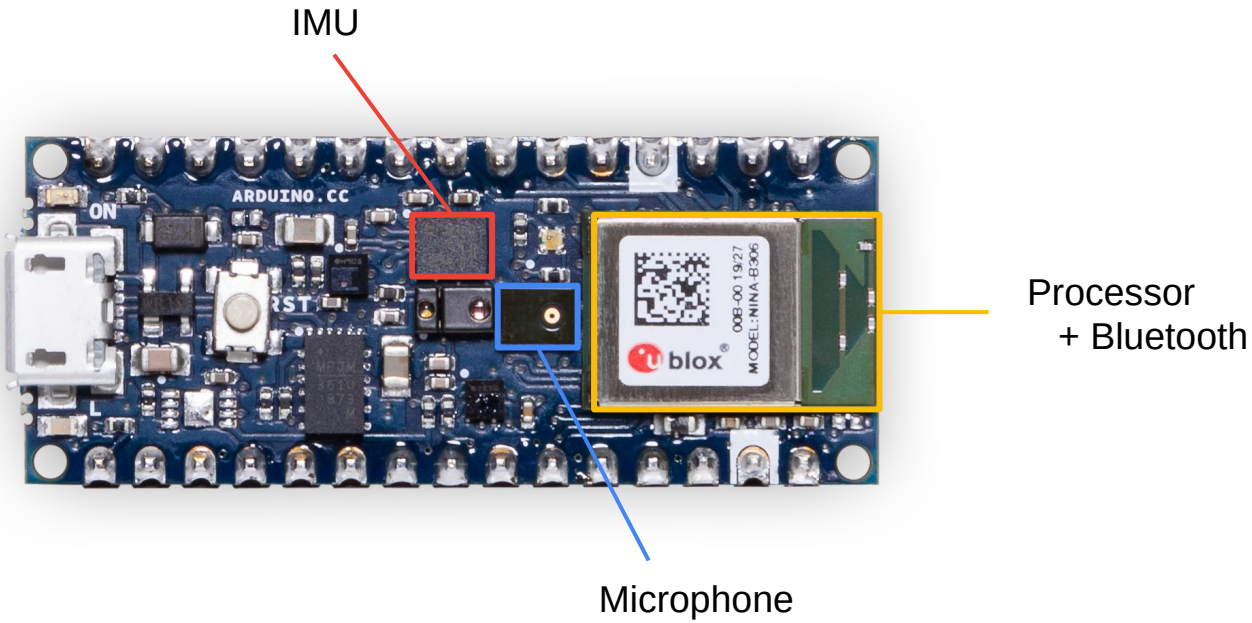
Developer Embedded Systems

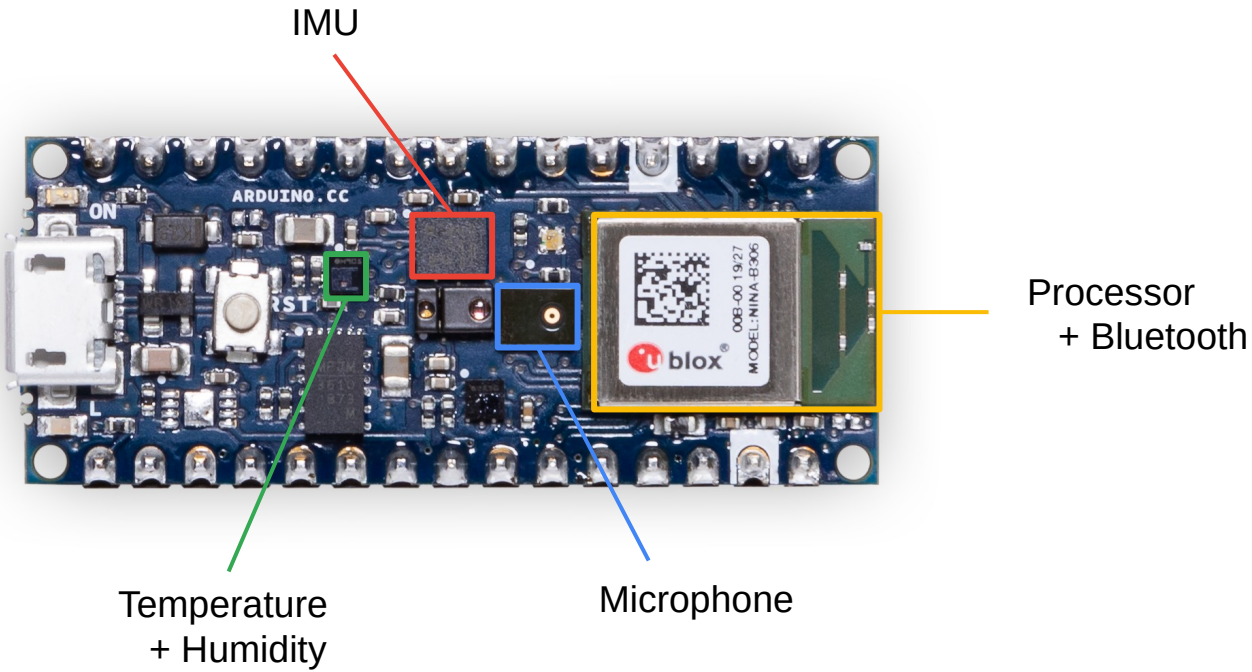


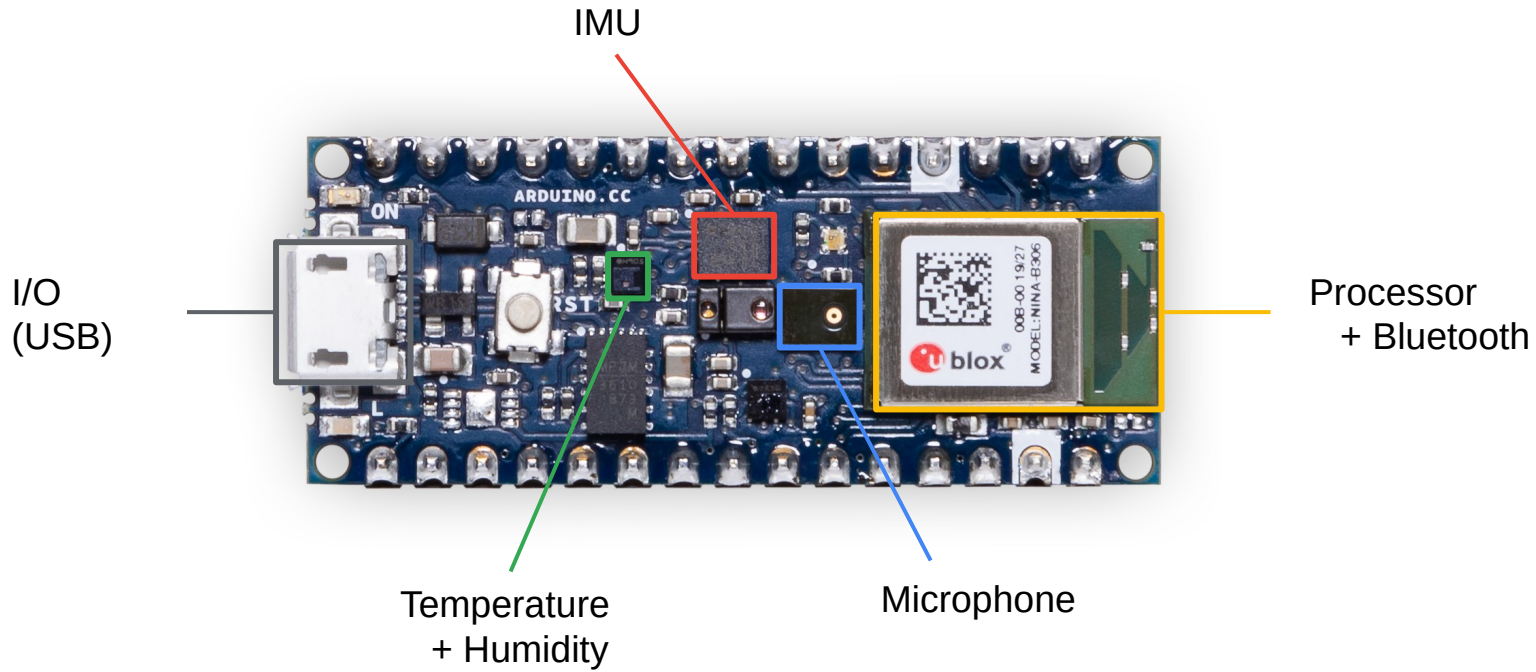


Processor
+ Bluetooth

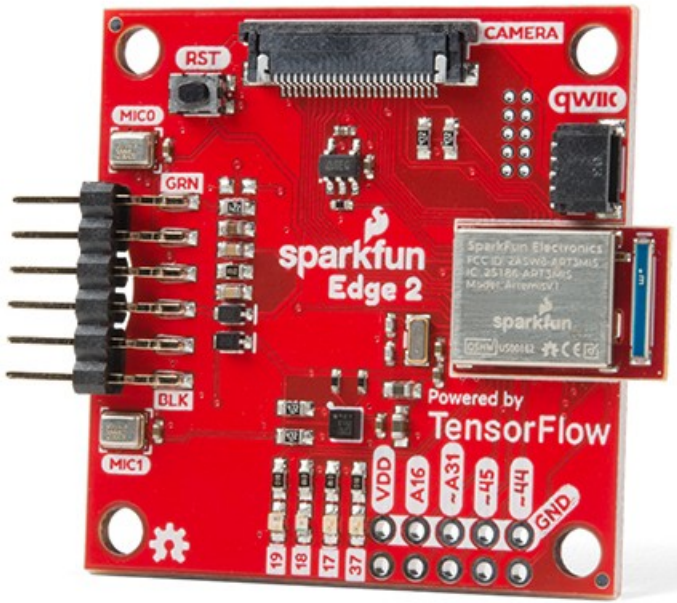

ARDUINO.CC
ON
blox
MODEL:NINA-B306
Processor
+ Bluetooth
Microphone

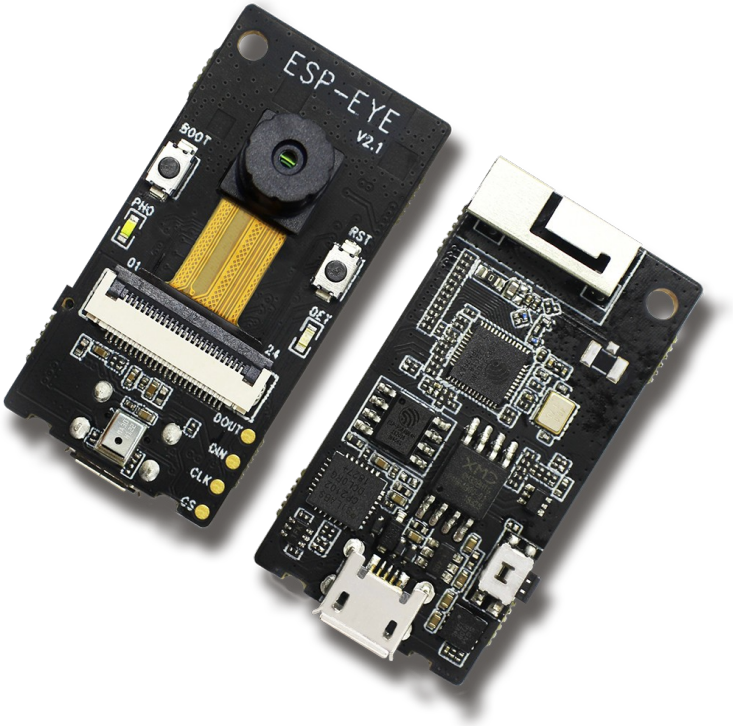

IMU
ARDUINO.CC
ON
RST
blox
MODEL:NINA-B306
Processor
+ Bluetooth
Microphone


IMU
ARDUINO.CC
ON
RST
blox
MODEL:NINA-B306
Processor
+ Bluetooth
Temperature
+ Humidity
Microphone


IMU
I/O
(USB)
Processor
+ Bluetooth
Temperature
+ Humidity
Microphone
ARDUINO.CC
ON
RST
blox
MODEL:NINA-B306

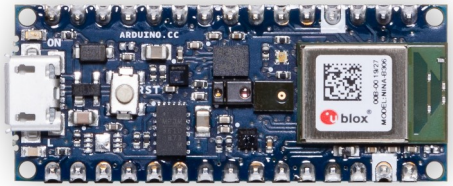

Himax
Reset

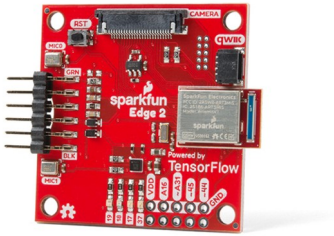

CAMERA
RST
MIC0
GRN
qwiic
sparkfun
Edge 2
BLK
MIC1
Powered by
TensorFlow
VDD
A16
~A31
~45
~44
GND
19
18
17
37


ESP-EYE
V2.1
BOOT
RST
01
24
DOUT
CLK

Embedded Systems









Embedded Systems

	Board	MCU / ASIC	Clock	Memory	Sensors	Radio
	Himax WE-I Plus EVB	HX6537-A 32-bit EM9D DSP	400 MHz	2MB flash 2MB RAM	Accelerometer, Mic, Camera	None
	Arduino Nano 33 BLE Sense	32-bit nRF52840	64 MHz	1MB flash 256kB RAM	Mic, IMU, Temp, Humidity, Gesture, Pressure, Proximity, Brightness, Color	BLE
	SparkFun Edge 2	32-bit ArtemisV1	48 MHz	1MB flash 384kB RAM	Accelerometer, Mic, Camera	BLE
	Espressif EYE	32-bit ESP32-D0WD	240 MHz	4MB flash 520kB RAM	Mic, Camera	WiFi, BLE

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Fullscreen
Show Presenter